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PROJECT PROPOSAL

**Research of Modern Methods of A/B
Testing on Samples from Interview Data**

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MOSCOW

2025

Abstract

A/B testing is a commonly used method for evaluating changes in systems; however, its effectiveness can be limited when applied to offline metrics. This limitation often occurs due to human error, which can reduce statistical power and increase the likelihood of false positives. This study aims to investigate whether incorporating individual variance into the analysis can improve the reliability of A/B tests. Specifically, this research will examine three different methods: (1) estimating variance directly from the sample data, (2) applying a fixed variance value according to the previous observations, and (3) integrating individual variance for each marker.

The hypothesis is that explicitly modeling individual variance will lead to more accurate statistical conclusions, enhance the power of the tests and reduce the occurrence of false positives.

In the future, this research could help explore how these methods perform with larger datasets.

Keywords

A/B Testing, variance estimation, offline metrics, false positives, statistical power, noise reduction.

Introduction

A/B testing is a common way to compare two versions of something, like a system or a product, to find out which one does better based on certain metrics. It's used a lot in areas such as digital marketing, product development, and software design. However, A/B testing has some big challenges when it comes to survey-based studies. In these cases, things like how participants behave inconsistently, personal biases, and the different ways people respond can create extra noise. This noise can make the statistical power of the test weaker and affect how reliable it is overall.

Traditional A/B testing methods often do not take into account the aforementioned problems, which can lead to a higher chance of false positives, which is when it seems like

the data point is significant but in reality it is not the case, which also makes it harder to select important results.

Objective

The objective of this study is to make comparisons of modern ways of estimating dispersion, with a particular emphasis on survey data.

Key Areas of Focus

1. The review of already existing methods

- The first central question is handling a review of current methods for estimating variance.

- Another crucial point is evaluating how well these approaches work with noise found in survey data.

2. Building an operational system

- The testing of various variance estimation methods will be compelling to create a structure, which will be using as artificially generated datasets as actual survey data for assessment.

3. Statistical Power and Error Rates

- It is also crucial to examine how these various approaches affect statistical power
- Besides, it is of vital importance to analyse how these methods influence error rates in A/B testing scenarios.

4. Identifying Best Practices

- Lastly, the current study will focus on determining optimal strategies to improve the precision and dependability of A/B testing using survey data.

Expected Contributions

Ultimately, this research aims to provide valuable insights intended to enhance the use of A/B testing in survey studies. By doing so, it will lead to more reliable and accurate experimental outcomes, which will ensure greater reliability and accuracy in findings.

Scientific Novelty and Practical Significance

This study stands out as it aims to improve how the variations in survey data are analyzed. A/B testing in its essence is commonly studied for online metrics, but there is a lack of the body of research when it comes to analyzing the respondents' subjective emotions or biased responses. The concentration on precise measure variance in this work will set the base for enhancing the trustworthiness of surveys in user experience (UX), social research, and machine learning.

From a practical standpoint, the outcomes of this research could be particularly applicable for both scholars in this field and professionals regularly using surveys in A/B testing. Getting a better handle on variance could lead to the creation of more intuitive surveys, thus simplifying the process of data interpretation, while at the same time improving its accuracy. This also holds significant implications in areas where people's biases may crucially distort the outcomes, in fields like customer satisfaction polls, political opinion surveys, or behavioural studies.

Definition of Key Terms

- A/B Testing – a statistical technique used to compare two versions of a variable in order to find out which functions on a higher level of accuracy.
- Variance Estimation – the method of measuring the variability within a sample or population, which is important for determining whether the results of an experiment are statistically significant.
- Statistical Power – the likelihood of accurately identifying a true effect in a hypothesis test.
- False Positive Rate (α) – the proportion of null hypotheses incorrectly rejected, leading to erroneous conclusions.
- Survey Data – the information collected from respondents through structured questionnaires, often subject to biases and measurement errors [9].

Literature review

The Significance of Variance

Variance estimation serves as a fundamental aspect of A/B testing, directly influencing hypothesis testing, confidence intervals, and statistical power. The work of Deng [2] critically analyzes the limitations of the delta method, particularly when observations exhibit dependence, which is a common characteristic in survey data. This complexity necessitates a review of alternative techniques aimed at variance reduction.

Challenges Specific to Survey Data in A/B Testing

The nature of survey data poses unique challenges that differ from simple numerical interactions. Subjective responses that depend on individual opinions introduce noise and variability, subsequently making interpretations more challenging to interpret. For instance, Nielsen Norman Group [6] examines the applicability of A/B testing in survey research, providing information on testing different survey formats and question designs.

Methodologies for Enhancing Offline A/B Testing

Several methodologies have been proposed to improve the accuracy and reliability of offline A/B testing. One approach involves the application of counterfactual inference techniques, which allow for the simulation of A/B tests using existing log data. Swaminathan et al. [3] provide a benchmark of these estimators by showing their correlation with business metrics observed in online A/B tests on commercial recommender systems. Additionally, Deng and Lu [7] discuss the importance of power analysis in experiments with clustered data and ratio metrics, emphasizing the need for specialized techniques to account for complex data structures. These methodologies aim to enhance the reliability of offline A/B testing by addressing inherent challenges such as data sparsity and individual variability [9].

Techniques for Variance Reduction

It is of utmost importance to improve sensitivity and reliability in A/B testing through variance reduction techniques. Research conducted by Microsoft has delved into the application of CUPED (Controlled Experiments Using Pre-Experiment Data) to revalue

variance estimation by integrating historical data [4]. The mentioned approach is particularly beneficial in survey experiments with the limitations of sample sizes.

Furthermore, Liu et al. [5] present a study focused on quantile-based A/B testing, which suggests a statistically valid and scalable methodology for experiments where skewed distributions may take place.

Network and Cluster Effects in Survey-Based Experiments

Traditional A/B testing methodologies usually make the assumption that observations are independent, hence the response of one participant does not affect the response of another. However, this assumption often falls short of understanding in the context of surveys where responses may be interrelated. In the work of Zhou et al [8], a new model known as cluster-adaptive network-based A/B testing, which considers the interconnections between respondents that exists in surveys. The application of adaptive randomisation presented in the article may balance the distribution of participants to treatment groups by taking into account clusters of interrelated responses.

Methods

The foundational methods of various estimation taken as a base of this study include:

1. Sample-based estimation – where variance is derived from the collected data straightforwardly.
2. Predefined total variance – where a constant variance is assigned to all data points.
3. Individual variance adjustment – where variance is computed independently in accordance with every marker taken.

The metrics used for evaluation are shown below:

1. Statistical power ($1 - \beta$) — which evaluates how precisely the true effects are identified during the test conduction.
2. False positive rate (α) — which monitors how often the null hypothesis is wrongly flagged in the process of the test.

3. P-value determines the level of statistical significance exhibited.

Results anticipated

The expected results are outlined down below:

Reduction of false positives

The first and foremost aspect of this study that plays a significant role is the maximum elimination of the false positives throughout the testing procedure. This phenomenon may occur due to the null hypothesis, which researchers often dismiss as being insignificant. The main struggle that happens during the A/B testing process is how to effectively distinguish real effects from random noise. The application of various estimates that is adjusted to each participant will overwhelmingly likely increase the statistical power, without the need to increase the size of samples.

The improvement of statistical power

The unaccounted noise in the data poses challenges to the final results. Such a result occurs when random fluctuations, being statistically insignificant data, is mistakenly interpreted as a pattern that is meaningful for the testing process. In turn, this leads to inaccurate predictions and falsely motivated decisions. Such occurrences may take place as the subject of research may be negatively or positively biased towards the questions or the testing method itself, and overemphasize minor aspects, thereby increasing the mistakes in self-reporting. By implementing individual various adjustments in this process, the false positive rates should be substantially reduced.

A wider application in the practical sense

There are certain fields where the algorithms of statistical approach to A/B testing are still being perfected — among them are behavioural testing, UX/UI research, and the research of client experience and consumer feedback. In the mentioned fields, the smaller samples of responses significantly skewed the final results, leading to improper conclusions and repercussions for the business results. In this case, individual adjustments may have a significant impact on the outcomes without imposing responsibilities on management and

researchers, while utilizing tools and methods already existing and successfully used in other fields.

Machine learning and data science usage

In conclusion, the methods proposed in this study may find further application in the rapidly developing field of machine learning and data science. The challenges that occur in the field are strongly connected with the inaccuracies of the data presented by a human being prone to biases and inconsistencies. The current study may address the existing gap by introducing modeling individual variables, which can stimulate further advancement in developing new, more sophisticated machine learning algorithms.

Conclusion

To conclude, the proposed study aimed at the improvement of reliability in A/B testing may come up with several possible ways to reduce the likelihood of the unaccounted factors such as human bias while responding to the survey questions, the noise in the data, and existing limitations in respective fields where this method is not yet researched. The solutions proposed rely on using newly developed A/B models, such as cluster-adaptive AI testing, new methods for experiments, and addressing existing challenges in the A/B testing itself.

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Word count 1665